```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def run_mattress_entropy_model():
    print("--- Loading Mattress Sales Mock Data ---")
    df = pd.read_csv("mattress_sales_mockup_3years.csv")
    df['Date'] = pd.to_datetime(df['Date'])

    # Aggregate weekly total sales
    weekly_sales = df.groupby('Date')['Units_Sold'].sum().reset_index()
    weekly_sales = weekly_sales.sort_values('Date').reset_index(drop=True)

    # Shannon Entropy Function
    def compute_shannon_entropy(series, bins=10):
        counts, _ = np.histogram(series, bins=bins, density=True)
        counts = counts[counts > 0]
        probs = counts / np.sum(counts)
        return -np.sum(probs * np.log2(probs))

    # Calculate 12-week rolling Shannon Entropy and Moving Averages
    window_size = 12
    rolling_entropy = []
    rolling_mean = []

    sales_values = weekly_sales['Units_Sold'].values

    for i in range(len(sales_values)):
        window_data = sales_values[:i+1] if i < window_size else sales_values[i-window_size+1:i+1]
        rolling_entropy.append(compute_shannon_entropy(window_data))
        rolling_mean.append(np.mean(window_data))

    weekly_sales['Entropy'] = rolling_entropy
    weekly_sales['Rolling_Mean'] = rolling_mean

    # Display console summary
    print(f"Total Weeks Analyzed: {len(weekly_sales)}")
    print(f"Latest Weekly Sales Volume: {weekly_sales['Units_Sold'].iloc[-1]} units")
    print(f"Latest 12-Week Rolling Shannon Entropy: {weekly_sales['Entropy'].iloc[-1]:.3f} bits")
    print("\n--- Q4 Sales Trend Prediction & Uncertainty Analysis ---")
    print("Interpretation: Lower entropy coupled with rising sales volume indicates predictable Q4 seasonal consolidation.")

    # Plotting Results
    fig, axes = plt.subplots(2, 1, figsize=(10, 8), sharex=True)

    axes[0].plot(weekly_sales['Date'], weekly_sales['Units_Sold'], label='Weekly Units Sold', color='#2b5c8f', alpha=0.6)
    axes[0].plot(weekly_sales['Date'], weekly_sales['Rolling_Mean'], label='12-Week Moving Average', color='#d95f02', linewidth=2)
    axes[0].set_ylabel('Mattress Unit Sales')
    axes[0].set_title('Mattress Category Sales Trend (3-Year Historical & Q4 Outlook)')
    axes[0].legend(loc='upper left')
    axes[0].grid(True, linestyle='--', alpha=0.5)

    axes[1].plot(weekly_sales['Date'], weekly_sales['Entropy'], label='Shannon Entropy (Bits)', color='#7570b3', linewidth=1.5)
    axes[1].axhline(y=weekly_sales['Entropy'].mean(), color='red', linestyle=':', label='Mean Entropy Baseline')
    axes[1].set_ylabel('Entropy (Bits)')
    axes[1].set_xlabel('Date')
    axes[1].set_title('Demand Uncertainty via Shannon Entropy')
    axes[1].legend(loc='upper left')
    axes[1].grid(True, linestyle='--', alpha=0.5)

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        plt.tight_layout()
        # Note: plt.show() requires a display environment; in headless mode it can be skipped or saved to file
        plt.savefig("mattress_entropy_forecast.png")
        print("Chart successfully saved to 'mattress_entropy_forecast.png'.")

if __name__ == '__main__':
    run_mattress_entropy_model()

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